

SAFETY STANDARD 安全标准	
UL certificate UL 证书	E341422
TUV certificate TUV 证书	R50220099
CQC certificate CQC 证书	CQC11002066606

COIL CHARACTERISTICS 线圈特性

Coil resistance 线圈电阻	103±10% Ω
Rated voltage 额定电压	12VDC
Max. allowable voltage 最大允许电压	130% of rated coil voltage
Rated power 额定功率	1.4W
Operate voltage 吸合电压	≥9.0VDC
Release voltage 释放电压	≤0.6VDC
Hold Voltage(See Notes 4) 保持电压	≥4.2VDC(@23℃) ≥5.4VDC(@85℃)

CONTACT RATINGS 触点规格

Contact configuration 触点结构	1 Form A (SPST)
Contact material 触点材料	Ag Alloy
Initial contact resistance 初始接触电阻	≤100mΩ at 6VDC/1A
Rated switching voltage 额定切换电压	277VAC
Rated current 额定电流	31A
Rated switching power 额定切换功率	8587VA
Minimum applicable load 最小使用负载	5VDC 100mA

Operate time 吸合时间	≤20ms, excluding bounce time ≤20ms, 不含触点抖动时间
Release time 释放时间	≤10ms, excluding bounce time ≤10ms, 不含触点抖动时间
Mechanical endurance 机械寿命	300K cycles, 180 cycles/minute
Electrical endurance (Resistive Load) 电气寿命（阻性负载）	31A, 277VAC, 30K cycles, 1s on/9s off 线圈通电 1 秒(12V/0.1 秒→下降 5.4~6.6V/0.9S) / 9 秒 OFF, (@85℃)
Contact Gap 接点间隙	2.05mm 以上
INSULATION PERFORMANCE 绝缘性能	
Dielectric strength 介电强度	2800VAC 1minute, between open contacts 2800VAC 1 分钟（断开触点间） 4,500VAC 1minute, between coil to contacts 4,500VAC 1 分钟（线圈与触点间）
Impulse withstand voltage 耐浪涌电压	10KV (1.2/50 μs), between coil to contacts 10KV（线圈与触点间）
Insulation resistance 绝缘电阻	1000MΩ at 500VDC, between open contacts and coil to contacts 1000MΩ（断开触点间及线圈与触点间） 155℃(F)
Insulation systems (UL) 绝缘系统	
Insulation type 绝缘类型	Basic insulation 基本绝缘
ENVIRONMENT PERFORMANCE 环境性能	
Category of protection (IEC61810-1) 密封类型	RT II (Flux Tight) 防助焊剂渗入型
Operating temperature 工作温度	-40~85℃
Operating humidity 工作湿度	20~85%RH
Storage temperature 储藏温度	-40~85℃
Storage humidity 储藏湿度	20~85%RH
Vibration resistance 耐振动	(1) Capability to function during vibration No opening or closing of any closed or opened contact circuit respectively exceed 10μs when the relay is subjected to vibration of 10~55Hz and 1.5mm

Shock resistance
耐冲击

dual amplitude in each of three mutually perpendicular axes for 10 minutes respectively, while it is in operate condition and in release condition.

抗误动作能力

动作/释放状态下，继电器在三个轴向耐受频率10~55Hz及振幅1.5mm的振动各10分钟，触点误动作不超过10微秒。

(2) Capability to function after vibration

No trouble on structure and characteristics after the relay is subjected to vibration of 10~55Hz and 1.5mm dual amplitude in each of three mutually perpendicular axes for 2 hours respectively.

振动耐久能力

继电器在三个轴向耐受振幅1.5mm及频率10~55Hz的振动各2小时，产品构造和性能无异常发生。

(1) Capability to function during shock

No opening or closing of any closed or opened contact circuit respectively exceed 10 μ s when the relay is subjected to shock of 98.1m/s² for 11ms in both directions of each of three mutually perpendicular axes for 3 times respectively, while it is in operate condition and in release condition.

抗误动作能力

动作/释放状态下，继电器在三轴六方向耐受加速度98.1m/s²及作用时间11毫秒的冲击各3次，触点误动作不超过10微秒。

(2) Capability to function after shock

No trouble on structure and characteristics after the relay is subjected to shock of 981m/s² for 6ms in both directions of each of three mutually perpendicular axes for 3 times respectively.

冲击耐久能力

继电器在三轴六方向耐受加速度981m/s²及作用时间6毫秒的冲击各3次，产品构造和性能无异常发生。

Cold resistance
耐低温

No trouble on structure and characteristics after placed at -40℃ for 240 hours and 2 hours recovery in standard atmospheric conditions.

-40℃中放置240小时并在标准大气条件中恢复2小时

Thermal resistance
 耐高温

后继电器构造和特性无异常。

No trouble on structure and characteristics after placed at 85℃ for 240 hours and 2 hours recovery in standard atmospheric conditions.

85℃中放置 240 小时并在标准大气条件中恢复 2 小时后继电器构造和特性无异常。

Humidity resistance
 耐湿度

No trouble on structure and characteristics after placed at 40℃&95%RH for 240 hours and 2 hours recovery in standard atmospheric conditions.

40℃及95%相对湿度中放置240小时并在标准大气条件中恢复2小时后继电器构造和特性无异常。

Thermal shock resistance
 耐冷热冲击

No trouble on structure and characteristics after endure 100 cycles of cyclic temperature and 2 hours recovery in standard atmospheric conditions, which the temperature cycle consists of -40℃ for 0.5 hour and 85℃ for 0.5 hour.

-40℃和85℃中各放置0.5小时为一个温度周期，循环100次，在标准大气条件中恢复2小时后继电器构造和特性无异常。

Terminal robustness
 引出端强度

No trouble on structure and characteristics after endure axial pushing/pulling force of 10N for 10 seconds.

继电器引出端承受 10 牛顿的轴向压入、拨出力，延时 10 秒，构造和性能无异常。

MARKING 产品标识

Position of marking
 标识位置

Side of relay cover

外壳侧面

Cover color
 外壳颜色

Black

黑色

Ink color
 字体颜色

Green

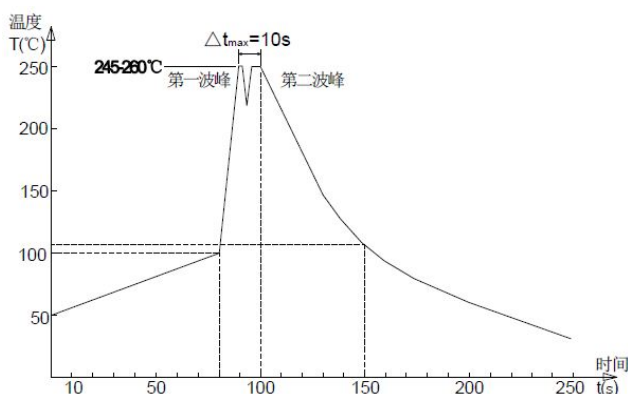
绿色

Trade mark
 商标



MOUNTING INFORMANTION 安装信息

Solderability 可焊性	260±5℃ for 5±0.5 seconds
Resistance to soldering heat 耐焊接热	270±5℃ for 10±1 seconds 350±10℃ for 3.5±0.5 seconds
Standard direction 标准方向	Relay PCB terminals downward 继电器 PCB 型引出端朝下
Terminals assignment and outline dimensions 引出端脚位和外形尺寸	Refer to APPENDIX 请参考附件
Wave soldering Condition 波峰焊条件	Refer the following Curve 参考下面曲线



ENGINEERING NOTES 注意事项

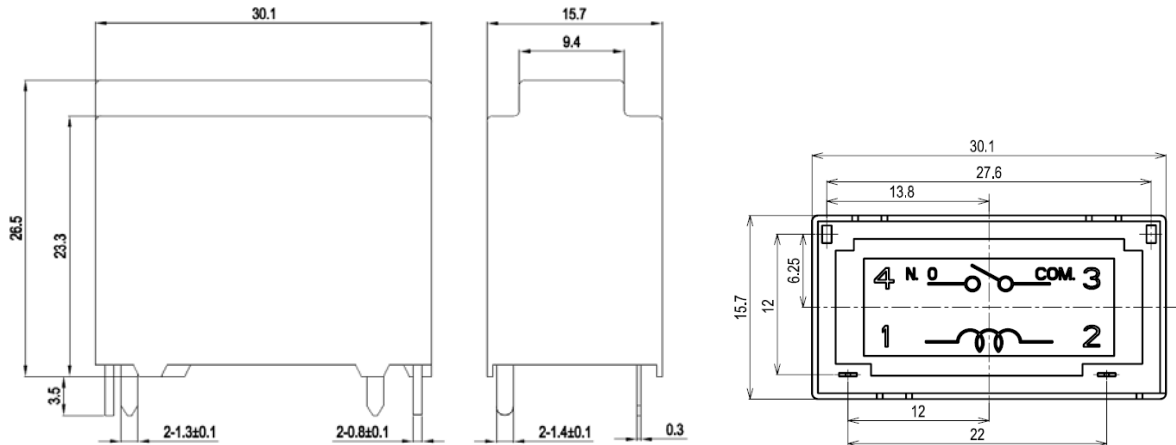
Unless otherwise explicitly stated, the standard environment conditions for measurement or testing are listed as followings:

除非特别申明，测量或试验的标准环境条件如下：

- (1) Ambient temperature is 23±5℃;
环境温度为 23±5℃;
- (2) Atmospheric pressure is 96±10% kPa;
大气压力为 96±10% kPa;
- (3) Relative humidity is 50%±25% RH.
相对湿度为 50%±25% RH.
- (4) When the ambient temperature > 23℃, coil voltage requires reduction to 5.4 to <6.6V after applying rated voltage for 100ms~200ms
当环境温度 > 23℃ 时，线圈施加额定电压 100ms~200ms 以后，电压需下降到 5.4~6.6V

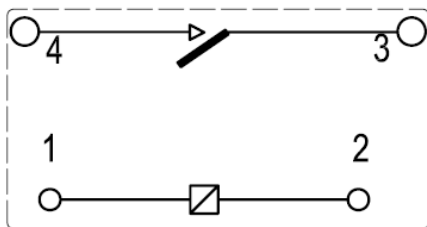
OUTLINE DIMENSION

外形尺寸图



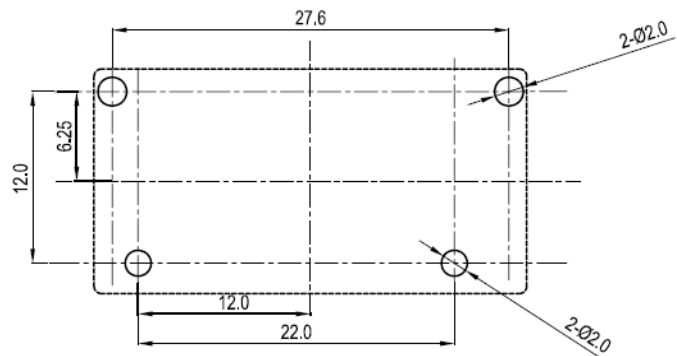
Wiring diagram (Bottom View)

接线图（底视图）



Mounting dimensions (Bottom View)

安装孔尺寸图（底视图）



Note: All unspecified tolerance (including outline dimension and PC board dimension) according to following listing

备注：产品外形尺寸未标注尺寸公差及 PC 板未注尺寸公差按下表执行。

产品外形尺寸未注尺寸公差 Outline dimensions hadn't specified tolerance		PC 板未注尺寸公差 PC board dimensions hadn't specified tolerance
外形尺寸 Outline dimensions	公差 Tolerance	± 0.1
≤ 0.3	± 0.1	
$< 0.3 \sim 1$	± 0.2	
$\leq 1 \sim 5$	± 0.3	
> 5	± 0.5	

Nomenclature

命名规则

	CHFN	-V	-1	12	H	A	2	F	(31A)
1. 产品系列 CHFN=PCB端子									
2. 密封型式 V: 透气孔外壳(耐助焊剂, RT II类) S: 密封外壳(耐清洗型, RT III类)									
3. 触点组数 1 = 一组触点									
4. 额定线圈电压 09 = 9VDC 12 = 12VDC 18 = 18VDC 24 = 24VDC									
5. 线圈功耗 H= 高容量型(1400mW)									
6. 触点构造 A= 常开型(SPST)									
7. 触点材质 2 = AgSnO ₂									
8. 绝缘系统 F& (空白) =Class F(155℃)									
9. 额定电流 (31A)=额定电流31A (35A)=额定电流35A									